

Quiz #4 Answer Sheet

CS 225 Data Structures

Spring 2026

7:00-8:00pm, Thursday, April 9

Name	Hanbin Zhou
ZJU ID	
Lab TA	

- This is a **closed book** and **closed notes** exam. No electronic aids are allowed, either.
- You should have 4 problems total on 3 pages. You must write on the exam.
- Unless otherwise stated in a problem, assume the best possible design of a particular implementation is being used.
- Unless the problem specifically says otherwise, (1) assume the code compiles, and thus any compiler error is an exam typo (though hopefully there are not any typos), and (2) assume you are NOT allowed to write any helper methods to help solve the problem, nor are you allowed to use additional arrays, lists, or other collection data structures unless we have said you can.
- Please put your name at the top of each page.
- Please fill in your **Lab Section** or your **Lab TA's name** to help us organize and grade your papers.
- Don't panic and good luck!

Problem 1 7 /10

Problem 2 10 /10

Problem 3 8 /10

Problem 4 9 /10

Total 34 /40



Problem 1 (10 points): Tree Basics and Traversal

1. d 2. d 3. Q
 4. (1) left, 47 (2) left, 18
 5.

```
void postorder(TreeNode* root) {
    if (root == nullptr) {
        return;
    }
```

```
    postorder(root->left);
    postorder(root->right);
    print (root);
}
```

我还没了,

print 定义没给,
 print (root->key)

或 std::cout << root->key.

Problem 2 (10 points): Binary Search Tree

Please write your answers clearly in the boxes below.

(a) BST validity check

[1] low

[2] root->key

[3] root->key

[4] high

(b) Find operation

[5] root

[6] root->right

(c) Delete operation

[7] root->right

[8] root->left

[9] root->left

[10] pred->key

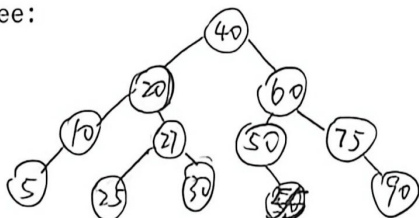


Problem 3 (10 points): AVL

1. 40-20-30-25 $O(\log n)$ $O(\log n)$

2. 30 LR

Tree:



3. 24 56

4. 6 51 10 50

RL 和 LR 看反了

Problem 4 (10 points): Huffman Tree & Running Time

1. (1) C

(2) ~~cd~~

没说哪个大哪个小

2. (1) b

(2)A. $O(n)$ B. $O(n)$ C. The order will change height. Running time for searching is $O(\log n)$

(3)

AVL Tree Deletion Step	Time Complexity (Worst)
1. find the node	$O(\log_2 n)$
2. delete the node like BST (left/right/2child, and pop)	$O(1)$
3. determine whether the tree after deletion is balanced	$O(\log_2 n)$
4. do rotation to make it balanced	$O(1)$
Totally	$O(\log_2 n)$

delete the node like BST

determine whether
If not, find the
unbalanced node